The Clebsch cubic
Recovering, orienting, and realizing — **III**

Arithmetic and harmonic realizations of the Clebsch cubic

Tavis Rudd

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From deep holes, the cubic takes shape, finds its bearings, and ***stands fixed while its shadows move****.*

Abstract

We determine the rational square class of Hitchin's degree-two incidence cover of harmonic cubics. Its function field is obtained by adjoining $\sqrt{5J_0}$. We define the Clebsch chart over $\mathbf{Q}(\sqrt{5})$, where $J_0 = 16\sigma_3^2$, and prove that the two configurations over the rational cubic xyz form the complete reduced fibre with residue algebra $\mathbf{Q}(\sqrt{5})$. The projective orthogonal transformation exchanging them has nonsquare spinor class after reduction modulo 11. On the golden Clebsch chart, normalization separates the pullback into two components. Choosing one component fixes a golden conference orientation and, through the primitive Petersen pair-sum map, the sign of the degree-six Clebsch cubic. Independently, the degree-six zonal harmonics on the ten icosahedral face axes contain a Clebsch four-space as their Petersen eigenspace, and their normalized spherical cubic restricts exactly as

$$\frac{1}{4\pi}\int_{S^2} F_y^3 = -\frac{784000}{1247103}\sigma_3(y).$$

The source comparison identifies orientation torsors, not the two cubics as polynomials: they live on different modules. No rational or integral identification between their ambient harmonic embeddings is asserted.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; spherical harmonics; Gaunt invariant.

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1 Introduction

Let

$$H = \ker\Big(\partial_x^2 + \partial_y^2 + \partial_z^2 : \mathbf{Q}[x,y,z]_3 \longrightarrow \mathbf{Q}[x,y,z]_1\Big)$$

be the rational seven-space of harmonic cubics for $Q(x,y,z) = x^2 + y^2 + z^2$. We use the polarization pairing on cubics normalized by

$$\langle p, (\, a \cdot x\,)^3 \rangle = p(a);$$

its restriction to H is nondegenerate. Hitchin associated to a general $[f] \in \mathbf{P}(H)$ two icosahedral six-axis configurations: equivalently, two Mukai–Umemura isotropic three-planes $U \subset H$ with $f \perp U$ [1, 2]. The incidence variety of pairs $([f], U)$ is therefore generically a degree-two cover of $\mathbf{P}(H)$. We determine its rational square class, which specifies the resulting quadratic function-field extension, and then study a separate occurrence of the same Clebsch invariant in degree-six spherical harmonics.

Let J_0 denote Hitchin's irreducible invariant sextic, with the normalization in his invariant calculation. Write

$$V = \{(y_1, \dots, y_5) : \textstyle\sum_i y_i = 0\}, \qquad \sigma_3(y) = \frac{1}{3}\textstyle\sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic. Here and below, $D(\sigma_3)$ is the principal open set $\{\sigma_3 \neq 0\}$. Because J_0 is a section of $\mathcal{O}_{\mathbf{P}(H)}(6)$, we use

$$K(\sqrt{cJ_0}), \qquad K = \mathbf{Q}(\mathbf{P}(H)),$$

as shorthand: choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$, put $j_s = J_0/s^2 \in K^\times$, and take $K(\sqrt{cj_s})$. Replacing s changes j_s by a square, so the quadratic extension is unchanged.

Theorem 1.1 (Arithmetic incidence cover)**.** *The function field of Hitchin's incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

For the Clebsch chart $\iota_t : \mathbf{P}(V_{\mathbf{Q}(\sqrt{5})}) \to \mathbf{P}(H_{\mathbf{Q}(\sqrt{5})})$ defined in Section 2,

$$\iota_t^* J_0 = 16\sigma_3^2.$$

The chart is not defined over $\mathbf{Q}$; Galois conjugation carries it to the chart of the conjugate icosahedron. The fibre over the rational point $[xyz]$ is nevertheless a complete reduced fibre with residue algebra $\mathbf{Q}(\sqrt{5})$.

The factor 5 has a concrete descent meaning: the cubic $[xyz]$ and its unordered geometric fibre are rational, but choosing either icosahedral configuration—equivalently, either conjugate Clebsch chart through that point—requires $\mathbf{Q}(\sqrt{5})$.

The choice can be retained globally after normalizing the pullback to the Clebsch chart. To state this precisely, orient the six displayed golden axes in their listed order and let C be their normalized conference matrix. Its switching-invariant triangle cubic is

$$Z_C(x) = \sum_{i<j<k} C_{ij}C_{jk}C_{ki}x_i x_j x_k.$$

It is translation-invariant and hence defines

$$Z_C \in \mathrm{Sym}^3((\mathbf{Q}^6/\mathbf{Q}\mathbf{1})^*).$$

The datum $[C, Z_C]_{\mathrm{or}}$ consists of the ordered six-axis marking, the orientation of its representative lattice, and the pair (C, Z_C), modulo simultaneous changes $a_i \mapsto d_i a_i$. Such a change sends C to DCD and fixes Z_C. The opposite source is represented by $(-C, -Z_C)$. The sign of the cubic generator is retained, so the brackets are not projective notation.

Theorem 1.2 (Common orientation source)**.** *Let $B = \mathbf{P}(V_E)$, where $E = \mathbf{Q}(\sqrt{5})$, and pull Hitchin's incidence normalization back along $\iota_t : B \to \mathbf{P}(H_E)$. The normalization of this pullback is a disjoint union*

$$B_+ \amalg B_-.$$

The fixed isotropic plane selects B_+, and incidence deck exchange interchanges the components. For an odd generator normalized on B_+, their equations are

$$z = 4\sqrt{5}\,\sigma_3, \qquad z = -4\sqrt{5}\,\sigma_3.$$

On $D(\sigma_3)$ they are the two distinct icosahedral configurations; on the Clebsch cubic the unnormalized branches meet, while the normalization remains disjoint.

Equip B_+ with the source $[C, Z_C]_{\mathrm{or}}$ supplied by the fixed plane and equip B_- with its deck-opposite. At $[xyz]$, the golden exchanger identifies this opposite with the conjugate configuration and the representative $(-C, -Z_C)$. Under the primitive pair-sum map

$$\beta : V \longrightarrow \mathbf{Q}^{\binom{5}{2}}, \qquad \beta(y)_{ij} = y_i + y_j,$$

the same choice fixes the linear lift of the Petersen four-space and hence the sign of the degree-six cubic in Theorem 1.3.

Here Z_C and σ_3 are different cubics on spaces of dimensions five and four. The theorem identifies the source of their orientations; it does not identify their polynomial domains.

Theorem 1.3 (Degree-six harmonic restriction)**.** *Let u_{ij}, $1 \leq i < j \leq 5$, be the explicitly labelled unit axes through opposite icosahedral faces in* (1)*, and set*

$$F_y(\omega) = \sum_{i<j} (y_i + y_j) P_6(u_{ij} \cdot \omega), \qquad P_6(s) = \frac{231s^6 - 315s^4 + 105s^2 - 5}{16}.$$

The coefficient vectors $(y_i + y_j)_{i<j}$ form the Petersen (-2)-eigenspace, and their zonal harmonics give an injective copy of V in the real degree-six spherical-harmonic space $\mathcal{H}_6$. With $\langle f, g\rangle = (4\pi)^{-1} \int_{S^2} fg$, the spherical cubic on this copy is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 \, d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

The arithmetic proof has three steps. Hitchin's degree and branch-sextic calculation reduces the generic extension to $K(\sqrt{cJ_0})$; the complete etale fibre at $[xyz]$, with residue algebra $\mathbf{Q}(\sqrt{5})$, determines $c = 5$; and the explicit projective orthogonal transformation between the two configurations yields spinor class [2]. Independently, the harmonic proof identifies the pair-sum coefficients with the Petersen (-2)-eigenspace and determines the invariant cubic from one exact spherical-moment calculation. Between these arguments, Theorem 1.2 normalizes the chart pullback, follows its deck involution through the golden conference matrix, and transports the resulting sign through the Petersen pair-sum map.

Hitchin proves the generic degree, identifies the branch sextic, constructs the Clebsch chart, computes its restriction, and classifies the fibre over xyz [1, §§2–4, 7–10] [2, §§4–5]. The arithmetic theorem fixes the remaining rational constant in that square class and computes the spinor class of the explicit specialization. The Mukai–Umemura threefold supplies the original threefold [4]; Hitchin supplies the incidence construction and every formula used here. Dye's finite theorem gives the complementary square-5 criterion for Clebsch hexagons [3]. The abstract equation $z^2 = 5J_{\mathbf{Z}}$ is etale on $D(10J_{\mathbf{Z}})$, but its comparison with the geometric incidence variety is proved only after inverting an unspecified integer.

The numerical harmonic restriction is logically independent of the incidence cover. The orientation-source theorem compares their signs only after choosing a sheet and a linear lift. It places the classical degree-six bond-order observable—the rotationally invariant cubic formed from degree-six harmonic coefficients— [6, 7] on the same four-space and computes its exact restriction. The coefficient $-784000/1247103$ has denominator divisible by 11, and this paper relates the two constructions through their four-dimensional A_5-module and invariant polynomial σ_3, rather than by a rational map between their ambient harmonic spaces.

2 The rational incidence cover

2.1 The Clebsch chart

Put $E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$, and let

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t], \\ [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}^2_E.$$

These are the six axes of the standard icosahedron. Define the linear subspace

$$V_t = \{p \in H_E : p(a) = 0 \text{ for every } [a] \in I_t\} \subset H_E.$$

This is the exact inclusion used below. Hitchin proves that V_t has dimension four and describes it as follows [1, §§2–4]. The rotation group of I_t permutes the five triples of mutually orthogonal symmetry planes. Let $q_1, \dots, q_5$ be the products of the three linear equations in those triples, with $q_1 = xyz$ and relative scalars chosen so that $\sum_i q_i = 0$. Then

$$\iota_t : V_E = \{(y_i) \in E^5 : \textstyle\sum_i y_i = 0\} \longrightarrow V_t, \qquad (y_i) \longmapsto \textstyle\sum_i y_i q_i$$

is an A_5-equivariant isomorphism. Thus the parameter of $xyz = q_1$ is

$$y^\circ = \frac{1}{5}(4, -1, -1, -1, -1), \qquad \sigma_3(y^\circ) = \frac{4}{25}.$$

With the polarization and sextic normalization fixed in the introduction, Hitchin's invariant calculation gives

$$\iota_t^* J_0 = 16\sigma_3^2, \qquad J_0(xyz) = \left(\frac{16}{25}\right)^2 [1, §10].$$

This chart is defined over E, not over $\mathbf{Q}$. The nontrivial automorphism $t \mapsto 1-t$ carries I_t and V_t to the distinct conjugate icosahedron and its vanishing space; the matrix R in Section 4 gives the corresponding transport. Hence ι_t is not a literal rational inclusion $V \hookrightarrow H$. The two displayed icosahedra have no common axis. Hitchin's two-icosahedron intersection theorem therefore gives

$$V_t \cap V_{1-t} = E \cdot xyz[1, \mathit{Proposition}\ 7].$$

Consequently the reduced union $\mathbf{P}(V_t) \cup \mathbf{P}(V_{1-t})$ descends to $\mathbf{Q}$: over E it is a pair of conjugate projective three-spaces meeting only at the rational point $[xyz]$. Denote the descended scheme by Z. Its normalization map is $\nu : Z^\nu \to Z$, where $Z^\nu = \mathbf{P}(V_t)$, regarded as a $\mathbf{Q}$-scheme through $\operatorname{Spec} E \to \operatorname{Spec} \mathbf{Q}$; after base change to E, this normalization is the disjoint union of the two conjugate components. The inverse image of the unique intersection point is

$$\nu^{-1}([xyz]) = \operatorname{Spec} E.$$

The local algebra makes the square class visible. Since the two linear branches are transverse, after base change to E and completion at $[xyz]$ we may write

$$\widehat{\mathcal{O}}_{Z,[xyz]} \otimes_{\mathbf{Q}} E = \frac{E[[x_1, x_2, x_3, y_1, y_2, y_3]]}{(x_i y_j : 1 \le i, j \le 3)}.$$

If $r = \sqrt{5}$, Galois conjugation exchanges x_i, y_i and sends r to $-r$. The invariant coordinates

$$a_i = x_i + y_i, \qquad b_i = r(x_i - y_i)$$

therefore give

$$\widehat{\mathcal{O}}_{Z,[xyz]} \cong \frac{\mathbf{Q}[[a_1,a_2,a_3,b_1,b_2,b_3]]}{(a_ib_j-a_jb_i,\ b_ib_j-5a_ia_j : 1\le i,j\le 3)}.$$

Its normalization is $E[[x_1,x_2,x_3]]$, with $a_i \mapsto x_i$ and $b_i \mapsto rx_i$. The conductor is the maximal ideal: every positive-degree series over E is generated by the images of the a_i, b_i, while the constants in the original ring are exactly $\mathbf{Q}$. Hence

$$\widetilde{\mathcal{O}}/\mathcal{O} \cong E/\mathbf{Q}$$

as $\mathbf{Q}$-vector spaces. Thus 5 is the splitting class of the two branches, not merely a value recovered from the sextic. This local model explains the descent geometry but does not replace the comparison with Hitchin's incidence scheme below. After base change to E, the pullback of the incidence double cover to $D(\sigma_3)$ is split. Galois conjugation exchanges both the two components and the two conjugate charts.

2.2 A rational incidence model

The function-field calculation requires a rational model, which can be given without descending either Clebsch chart. Let $W = \mathbf{Q}^3$ with the quadratic form Q from the introduction. For $a \in W$, infinitesimal rotation about a acts on H; put

$$\omega_a(p,q) = \langle a\cdot p, q\rangle.$$

These three rational skew forms define a closed subscheme

$$X = \{U \in \mathrm{Gr}(3,H) : \omega_a|_U = 0 \text{ for every } a \in W\}.$$

Hitchin identifies $X_{\mathbf{C}}$ with the Mukai–Umemura threefold [2, §§4–5]; in particular X is geometrically integral and smooth. The incidence scheme used here is

$$\mathcal{I} = \{([f],U) \in \mathbf{P}(H)\times X : \langle f,U\rangle = 0\}.$$

Let $\pi : \mathcal{I} \to \mathbf{P}(H)$ be the first projection. It is a projective bundle over X, is therefore normal and integral, and all its displayed equations are defined over $\mathbf{Q}$. Mukai and Umemura introduced the underlying threefold; Hitchin supplies this Grassmannian model and the incidence calculation [4, 2].

There is also a canonical link from the chart model to the finite incidence cover. Let

$$\mathcal{N} = \mathrm{Spec}_{\mathbf{P}(H)}\, \pi_*\mathcal{O}_{\mathcal{I}}.$$

Stein factorization gives $\mathcal{I} \to \mathcal{N} \to \mathbf{P}(H)$. Since $\mathcal{I}$ is normal and the degree calculation below makes its generic field quadratic over $\mathbf{Q}(\mathbf{P}(H))$, $\mathcal{N}$ is the normalization of $\mathbf{P}(H)$ in that field. For the icosahedron I_t, let $U_t \in X(E)$ be Hitchin's isotropic three-plane. The polarization identity and Hitchin's description give [1, §§2–5]

$$V_t = U_t^{\perp},$$

so $[p] \mapsto ([p],U_t)$ defines a section $\mathbf{P}(V_t) \to \mathcal{I}_E$. The two conjugate sections descend to a morphism

$$Z^\nu \longrightarrow \mathcal{I} \longrightarrow \mathcal{N}$$

over $\mathbf{P}(H)$. Over the finite etale neighborhood of $[xyz]$ constructed below, the local comparison proves that this morphism carries $\nu^{-1}([xyz]) = \mathrm{Spec}\, E$ isomorphically onto the golden incidence fibre.

2.3 Proof of Theorem 1.1

Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension.

Square class from the branch divisor. Write $K = \mathbf{Q}(\mathbf{P}(H))$. In characteristic zero every quadratic extension of K has the form $K(\sqrt{d})$. The prime divisors at which d has odd valuation are exactly the branch divisors. Hitchin identifies the branch hypersurface with the irreducible invariant sextic $J_0 = 0$ [1, §§9–10].

Choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$ and set

$$j_s = \frac{J_0}{s^2} \in K^\times.$$

Its odd valuations occur exactly along $J_0 = 0$, because the divisor of s^2 is even. Consequently

$$\mathrm{div}(d/j_s) = 2D$$

for a divisor D on $\mathbf{P}(H)$. The class of D is two-torsion. Since $\mathrm{Pic}(\mathbf{P}(H)) = \mathbf{Z}$, D is principal. Thus

$$d = c j_s g^2$$

for some $g \in K^\times$ and a unique $c \in \mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. The incidence extension is therefore $K(\sqrt{cJ_0})$.

The local comparison at xyz. Hitchin's classification gives exactly the two distinct configurations I_t and I_{1-t} over $[xyz]$ [1, §§3, 7–9]. Thus π is quasi-finite over this point. Because π is proper, there is a Zariski neighborhood B of $[xyz]$ on which π is finite. Let $\widetilde{B}$ be the normalization of B in the quadratic generic field $K(\sqrt{cJ_0})$. The normal finite scheme $\pi^{-1}(B)$ is birational to $\widetilde{B}$, hence is isomorphic to it. Hitchin's branch calculation identifies the branch locus with $J_0 = 0$. Since $J_0(xyz) \neq 0$, shrinking B away from this divisor makes the finite normalization etale. The two distinct geometric points therefore form the complete reduced scheme-theoretic fibre, whose residue algebra is

$$E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5}).$$

Choose a local generator e of $\mathcal{O}_{\mathbf{P}(H)}(3)$ at $[xyz]$, and write $j_e = J_0/e^2$. The normalization is locally $w^2 = cj_e$. Changing e multiplies j_e by a unit square. Moreover, $j_e(xyz)$ differs from the displayed value $J_0(xyz) = (16/25)^2$ by a rational square. Specialization therefore gives the residue algebra $\mathbf{Q}(\sqrt{c})$. Comparison with the preceding fibre proves $c = 5$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. This also explains why evaluating the generic square class at the point is legitimate: the point lies in the finite etale locus and the change from j_s to j_e is a unit square there.

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \qquad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 1.1. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N, but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

3 The common orientation source

This section keeps track of the sign lost by projectivizing the Clebsch chart. It first normalizes the pulled-back incidence cover, then follows deck exchange through the golden six-axis data and the Petersen coefficient module.

3.1 Normalization over the Clebsch chart

Let $\mathcal{N} \to \mathbf{P}(H)$ be the incidence normalization from Section 2, and put $B = \mathbf{P}(V_E)$. On the chart, Theorem 1.1 and Hitchin's restriction identity give

$$z^2 = 5\iota_t^* J_0 = 80\sigma_3^2 = (z - 4\sqrt{5}\,\sigma_3)(z + 4\sqrt{5}\,\sigma_3).$$

Each factor has function field $E(B)$, and B is normal. The normalization of $B \times_{\mathbf{P}(H)} \mathcal{N}_E$ is therefore two copies $B_+ \amalg B_-$. The constant plane U_t supplies one component; deck exchange supplies the other. On $D(\sigma_3)$, the two linear factors differ by a unit, so the original pullback is already their product and is finite etale. Hitchin's chart classification identifies this as exactly the locus of two distinct regular configurations. Along $\sigma_3 = 0$, the factors meet before normalization, but their normal components remain separate.

This factorization is an equality of sections after the same local trivialization of $\mathcal{O}(3)$ used in the square-class proof. It does not strengthen the integral statement: comparison with the geometric incidence model still holds only over an unspecified $\mathbf{Z}[1/N]$.

3.2 The golden involution

For the ordered representatives $a_0, \dots, a_5$ of I_t, their Gram matrix has the form

$$G = (t+2)I + tC, \qquad C = \begin{pmatrix} 0 & 1 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & 1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 1 & -1 & 0 \end{pmatrix}.$$

Direct summation of the six rank-one matrices gives $\sum_i a_i a_i^{\mathsf{T}} = 2(t+2)I_3$, hence $G^2 = 2(t+2)G$. Substituting $G = (t+2)I + tC$ and using $(t+2)^2 = 5t^2$ gives $C^2 = 5I$. Changing signs of axis representatives switches C to DCD, but every triangle product and hence Z_C is unchanged. The off-diagonal entries of C^2 also give

$$\sum_{k \neq i,j} C_{ij} C_{jk} C_{ki} = 0.$$

Thus the directional derivative of Z_C along $\mathbf{1}$ vanishes, so $Z_C(x + u\mathbf{1}) = Z_C(x)$ and the cubic descends to $\mathbf{Q}^6/\mathbf{Q}\mathbf{1}$.

Let a_i' denote the conjugate axes, obtained by $t \mapsto 1-t$, and let R be the exchanger of Section 4. Direct substitution gives

$$Ra_i = td_i a'_{p(i)}, \qquad p = (0,1,5,4,2,3), \qquad (d_i) = (1,-1,1,1,1,1).$$

Using $t^2(1-t) = -t$, one obtains

$$d_i d_j C_{p(i)p(j)} = -C_{ij}.$$

Thus, at the golden fibre, conjugation transported by R changes C to $-C$ and Z_C to $-Z_C$. The lone negative d_i is only a change of axis representatives and does not cause the reversal; switching alone leaves the triangle cubic fixed.

No global marking of the varying configurations on B_- is needed or asserted. The fixed section U_t marks B_+ by the constant source $[C, Z_C]_{\mathrm{or}}$. We mark B_- by its deck-opposite, meaning that the normalized odd generator changes sign. The calculation at $[xyz]$ proves that this algebraic convention agrees with the actual conjugate golden configuration there; it is not used to infer a pointwise conference labeling elsewhere. Since the normalized components remain disjoint above $\sigma_3 = 0$, the two source labels extend there even though the literal two-configuration interpretation fails.

3.3 Transport to degree six

Write Hitchin's chart as $\iota_t(y) = \sum_i y_i q_i^{(t)}$, with $q_1 = xyz$. Since $R(xyz) = -xyz$, equivariance and irreducibility of the four-module give

$$Rq_i^{(t)} = -q_i^{(1-t)}$$

after transporting the five plane-triple labels. Normalize the lift on B_+ by $q_1 = xyz$; the deck-opposite source carries the lift $-\iota_t$. The displayed exchanger calculation shows that this normalization agrees with golden conjugation at $[xyz]$. Thus the lift is part of the oriented source datum, rather than a purported global marking of the second family.

The primitive integral map

$$\beta(y)_{ij} = y_i + y_j$$

lands in the Petersen (-2)-eigenspace and satisfies

$$\sum_{i<j} (y_i + y_j)^2 = 3 \sum_i y_i^2, \qquad y_i = \frac{1}{3} \sum_{j \neq i} \beta(y)_{ij}.$$

Multiplicity one makes this the canonical comparison after the five labels are fixed. Reversing the chart lift reverses y, the zonal harmonic F_y, and its cubic integral. The two signs in the normalized incidence cover therefore select the two signs of the exact Gaunt cubic. This proves Theorem 1.2.

There is no ambient linear comparison between harmonic degrees three and six: their irreducible rotation modules have dimensions seven and thirteen. The comparison exists only on the chosen Clebsch four-space, and it stops at the oriented source class $[C, Z_C]_{\mathrm{or}}$.

4 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz. Its two icosahedral six-axis configurations are Galois conjugate over $\mathbf{Q}(\sqrt{5})$. The exchanger is the projective orthogonal transformation carrying one configuration to the other. Over a finite field where the configurations become rational, it has spinor class [2]; modulo 11 this class is nontrivial.

4.1 The golden fibre

Retain the field E, parameter t, and six-set I_t from Section 2. These six projective points are its axes through opposite vertices. Thus an icosahedral configuration here means the unordered six-set I_t. A point of the incidence cover records one chosen six-set; its two points over xyz correspond to the two conjugate configurations.

Proposition 4.1 (Golden fibre)**.** *The points of I_t lie on $xyz = 0$, have a common nonzero norm, have normalized squared inner product $1/5$, and form a six-arc. The two configurations are conjugate geometric configurations defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's incidence cover, whose point records the chosen six-set, they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$, while

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point determinants are $\pm 2t$ or $\pm 2t^2$, so none vanishes under the stated hypotheses. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. □

The coordinate conclusions in the proposition remain valid after reduction to any field of characteristic different from 2 and 5, once a root of $t^2 - t - 1$ is chosen. This does not extend the complete-fibre assertion: the comparison with Hitchin's geometric incidence cover is available in characteristic zero and, after spreading out, only away from an unspecified finite set of primes.

4.2 The exchanger

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \qquad R^2 = \operatorname{diag}(1, -1, -1), \qquad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two configurations.

4.3 Finite-field specialization

Let k be a finite field of odd characteristic other than 5, and suppose that 5 is a square in k. Then $t^2 - t - 1$ has two distinct roots in k, so the displayed equations give two rational configurations exchanged by R.

Proposition 4.2 (Spinor specialization)**.** *The spinor class of R is $[2] \in k^\times/k^{\times 2}$. Under the standard identifications*

$$\mathrm{SO}_3(k)/\Omega_3(k) \cong \mathrm{PGL}_2(k)/\operatorname{PSL}_2(k),$$

the exchanger lies outside $\mathrm{PSL}_2(k)$ exactly when 2 is nonsquare. In particular, over $\mathbf{F}_{11}$ it represents the nontrivial element of

$$T_{11} = \mathrm{PGL}_2(11)/\operatorname{PSL}_2(11).$$

Proof. For $Q(x,y,z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz-plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } k^\times/k^{\times 2}.$$

The standard identifications

$$\mathrm{SO}_3(k) \cong \mathrm{PGL}_2(k), \qquad \Omega_3(k) \cong \mathrm{PSL}_2(k)$$

identify the spinor quotient with the projective quotient. Finally, the roots modulo 11 are 4 and 8, and 2 is nonsquare modulo 11, proving the specialization. ☐

Thus two independent square classes govern the finite specialization: [5] determines whether the two displayed configurations are rational, while [2] determines whether their exchanger lies in the special projective subgroup. Dye's existence criterion for Clebsch hexagons has the same [5] boundary [3, Theorem 1].

Integral boundary. Over a finite field of characteristic different from 2 and 5, the fibre $\mathcal{Q}_f$ of the abstract quadratic cover $z^2 = 5J(f)$ over a rational point f gives

$$\#\mathcal{Q}_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R; we do not replace the finite exceptional set by $p \neq 2, 5$.

5 The degree-six harmonic realization

Here $\mathcal{H}_6$ denotes the real degree-six spherical harmonics on S^2. The construction passes from icosahedral face axes to the Petersen coefficient module and then to $\mathcal{H}_6$.

Put $\phi = (1+\sqrt{5})/2$. The following vectors have squared norm 3; their projective classes are the ten axes through opposite faces of an icosahedron:

$$\begin{aligned} &v_{12} = (-1,-1,-1), && v_{13} = (-1,-1,1), && v_{14} = (-1,1,-1), && v_{15} = (-1,1,1), \\ &v_{34} = (0,-\phi^{-1},-\phi), && v_{25} = (0,-\phi^{-1},\phi), && v_{45} = (-\phi^{-1},-\phi,0), && \\ &v_{23} = (-\phi^{-1},\phi,0), && v_{35} = (-\phi,0,-\phi^{-1}), && v_{24} = (\phi,0,-\phi^{-1}). && \end{aligned} \tag{1}$$

We use the unit representatives $u_{ij} = v_{ij}/\sqrt{3}$. Two labels are adjacent when the pairs are disjoint. This is the Petersen graph $KG(5,2)$, and the labeling is geometric because, for distinct axes,

$$(u_{ij} \cdot u_{kl})^2 = \begin{cases} 5/9, & \{i,j\} \cap \{k,l\} = \varnothing, \\ 1/9, & \text{otherwise.} \end{cases}$$

Replacing ϕ by its conjugate gives the conjugate embedded configuration. The abstract Petersen incidence and the formulas below are unchanged after the corresponding relabeling. We do not claim that this displayed embedding of its four-space is defined over $\mathbf{Q}$.

The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \qquad L(a)(\omega) = \sum_{i<j} a_{ij} P_6(u_{ij} \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16}\left(231s^6 - 315s^4 + 105s^2 - 5\right).$$

We equip $\mathcal{H}_6$ with the normalized spherical inner product

$$\langle f, g\rangle = \frac{1}{4\pi}\int_{S^2} f(\omega)g(\omega)\,d\omega.$$

5.1 Proof of Theorem 1.3

Let A be the Petersen adjacency matrix in the labeling (1). Direct evaluation gives the reproducing-kernel matrix

$$K_{ef} = P_6(u_e \cdot u_f) = \frac{1}{243}(196I + 47J - 112A).$$

The addition theorem for spherical harmonics gives

$$G_{ef} = \langle P_6(u_e \cdot -), P_6(u_f \cdot -)\rangle = \frac{1}{13}K_{ef}.$$

Thus G, not K, is the Gram matrix for the stated inner product. Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{1053}, \qquad \frac{140}{1053}, \qquad \frac{28}{1053}.$$

They are positive, so L is injective.

The Petersen eigenvalues are $3, -2, 1$ on $\mathbf{1}, V_4, V_5$. For $a_{ij} = y_i + y_j$ with $\sum_i y_i = 0$,

$$(Aa)_{ij} = \sum_{\{k,l\}\cap\{i,j\}=\varnothing} (y_k + y_l) = -2(y_i + y_j).$$

These vectors form the four-dimensional (-2)-eigenspace.

This identifies the abstract bridge uniquely. The action of A_5 on five letters is two-transitive, so an endomorphism of $\mathbf{Q}^5$ commuting with A_5 has one constant diagonal entry and one constant off-diagonal entry. Its restriction to V is therefore scalar. Consequently, once the five plane triples are identified, every A_5-equivariant comparison between the arithmetic coordinate module and the Petersen coefficient module is a scalar multiple of

$$y \longmapsto (y_i + y_j)_{i<j}.$$

The inverse on its image is

$$y_i = \frac{1}{3}\sum_{j\neq i} a_{ij}.$$

Requiring the comparison to preserve σ_3 forces the scalar to have cube one, hence to equal one over $\mathbf{Q}(\sqrt{5})$. The abstract comparison is therefore canonical with the stated normalization. It still supplies no map between the two ambient harmonic spaces. The linear lift selected in Theorem 1.2 fixes its sign.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi}\int F_y^2 = \frac{2800}{351}, \qquad \frac{1}{4\pi}\int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the coefficient in Theorem 1.3. The moments follow by expanding P_6 and applying

$$\frac{1}{4\pi}\int_{S^2} \omega_1^{2a}\omega_2^{2b}\omega_3^{2c}\,d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

monomials with an odd exponent integrate to zero.

Remark 5.1. In the orthonormal Condon–Shortley convention, write $F = \sum_{m=-6}^{6} q_{6m}Y_{6m}$ and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1}q_{6m_2}q_{6m_3}.$$

The Gaunt formula and $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right) = -20/\sqrt{46189}$ give

$$\int_{S^2} F^3\,d\omega = -\frac{130}{\sqrt{3553\pi}}\,W_6(F).$$

This is the standard unnormalized degree-six bond-order invariant [6, 7, 8]. Combining this conversion with Theorem 1.3 gives the exact restriction in that standard language:

$$W_6(F_y) = \frac{313600\pi^{3/2}}{4563\sqrt{3553}}\,\sigma_3(y).$$

6 Reproducibility and data availability

The square-class argument and both specialization propositions are proved in the text from Hitchin's incidence calculation and the displayed equations. A paper-local exact-arithmetic bundle independently checks the golden configurations, the exchanger, and its spinor decomposition.

A second paper-local bundle reconstructs the ten axes over $\mathbf{Q}(\sqrt{5})$, distinguishes the reproducing kernel from the normalized spherical Gram matrix, verifies the Petersen labeling, and evaluates the quadratic and cubic moments. Its independent replay uses a separate dense polynomial representation and checks the Wigner $3j$ conversion. Neither calculation uses floating-point arithmetic.

A third bundle checks the conference square, the golden exchanger and its transport of C and Z_C, the scalar factorization of the pulled-back cover, and the primitive Petersen pair-sum identities. It reuses the canonical harmonic certificate only through its pinned digest and exact witness values. Normalization of the pulled-back incidence scheme and its extension across the branch divisor remain human arguments.

The `verification` directory distributed with the paper contains the exact programs, canonical certificates, checksums, toolchain requirements, and aggregate replay command. It also records the exact claim-to-evidence correspondence. The manifest claims no Lean coverage. A supplemental formal package proves the displayed conference square, switching invariance, triangle reversal, and pair-sum mechanisms, but no manuscript theorem depends on it and no new formal surface is needed for the scheme-theoretic normalization. The incidence model, square class, local fibre comparison, spinor specialization, face-axis geometry, and spherical moments are not formal dependencies. These exact-arithmetic audits are independent checks rather than steps in the proofs.

7 Conclusion

Hitchin's two conjugate Clebsch charts live over $\mathbf{Q}(\sqrt{5})$, where the branch sextic becomes $16\sigma_3^2$. Their common rational point $[xyz]$ lies in the finite etale locus of the rational incidence model. Their descended union has a nonsplit two-branch singularity there; its normalization fibre and conductor recover $\mathbf{Q}(\sqrt{5})/\mathbf{Q}$. The fixed-configuration sections map this normalization model canonically to the incidence normalization, whose complete reduced fibre determines the remaining rational square class. The explicit configurations also have a well-defined specialization at 11, without requiring a global integral incidence model there.

The degree-six face-axis construction gives a second exact role for σ_3: it is the restriction of the normalized spherical cubic to the Petersen four-space. The two constructions use isomorphic Clebsch modules and the same invariant polynomial. Normalizing the incidence pullback retains the choice of sheet across the Clebsch cubic, even where the unnormalized branches meet. The fixed sheet carries that choice to $[C, Z_C]_{\mathrm{or}}$; deck exchange gives its opposite, and the golden exchanger identifies the opposite with the conjugate configuration at $[xyz]$. The primitive Petersen map carries its sign to the degree-six Gaunt cubic. This is a comparison of orientation sources, not an equality between Z_C and σ_3 or a map between their ambient harmonic spaces. Determining the exact finite set of primes that must be inverted for the geometric incidence comparison remains a concrete arithmetic problem.

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References

[1] N. Hitchin, *Spherical harmonics and the icosahedron*, CRM Proceedings and Lecture Notes 47 (2009), 215–231; doi:10.1090/crmp/047/14.

[2] N. Hitchin, *Vector bundles and the icosahedron*, Contemporary Mathematics 522 (2010), 71–87; doi:10.1090/conm/522/10292.

[3] R. H. Dye, *Hexagons, conics, A_5 and* $\mathrm{PSL}_2(K)$, Journal of the London Mathematical Society (2) 44 (1991), 270–286; doi:10.1112/jlms/s2-44.2.270.

[4] S. Mukai and H. Umemura, *Minimal rational threefolds*, in *Algebraic Geometry (Tokyo/Kyoto, 1982)*, Lecture Notes in Mathematics 1016, Springer (1983), 490–518; doi:10.1007/BFb0099976.

[5] V. Krishnamoorthy, T. Shaska, and H. Völklein, *Invariants of binary forms*, arXiv:1209.0446.

[6] P. J. Steinhardt, D. R. Nelson, and M. Ronchetti, *Icosahedral bond orientational order in supercooled liquids*, Physical Review Letters **47** (1981), 1297–1300; doi:10.1103/PhysRevLett.47.1297.

[7] P. J. Steinhardt, D. R. Nelson, and M. Ronchetti, *Bond-orientational order in liquids and glasses*, Physical Review B **28** (1983), 784–805; doi:10.1103/PhysRevB.28.784.

[8] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Release 1.2.7 (2026), §§14.30 and 34.3; dlmf.nist.gov.